



IIT ROORKEE



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Measuring Intra-Industry Trade - II

Dr. PRATAP C. MOHANTY

DEPT. OF HUMANITIES & SOCIAL SCIENCES, IIT ROORKEE



THE CONTEXT

- IIT constitutes a major portion of a country's trade
- Empirical evidences since the 1970s reveal that a large part of world trade takes place between similar countries and products (Grubel and Lloyd 1975), which cannot be explained by the inter-industry trade theories.
- Trade in varieties of a product characterized by different attributes is **HIIT** and when it is characterized by quality difference, such trade is **VIIT**.
- HIIT - consumers view goods as a bundle of characteristics
- VIIT - A higher-end processor-enabled personal computer may be an example of such a variation in quality.
- The reasons for quality differentiation are income disparity across and within countries, technological disadvantage and difference in endowment of skilled labour and human capital.
- The horizontal models are considered to be more relevant for understanding the occurrence of IIT among the developed countries and the vertical models for explaining IIT between unequal partners.

- North-South trade differences
- developing countries are producers and exporters of poor quality products (Acharyya 2005; Marjit and Raychaudhuri 1997)
- Recently, the developing countries suffer further due to non-tariff measures by developed countries
- Hausmann et al (2007) is that “countries become what they produce”.
- Since quality variation depends on technological advancement and the availability of skilled labour, the transition of IIT from horizontal to vertical or the other way round can have important policy implications for the domestic economy.
- the share of intermediate goods dominate IIT globally (Brulhart 2008; Marini 2017)

- The Indian experience is interesting to analyse because even though India is an unskilled-labour abundant country, its export even to the skilled-labour abundant countries has a large share of high-technology and skill-intensive commodities (Acharyya 2013)
- productivity of the export basket at the cross-country level, China and India were pointed out to be outliers in the set of developing countries (Rodrik 2006)
- This result was obtained without including software exports by India. Hence, Rodrik (2006) expected India's EXPY (which measures the productivity of export basket) to be even higher with software exports.
- there is hardly any systematic analysis to decompose India's total IIT into horizontal and vertical trade excepting the studies by Bagchi et al (2014) for six Indian industries over 1990–2013 and Srivastava and Medury (2011) during 2000–08.

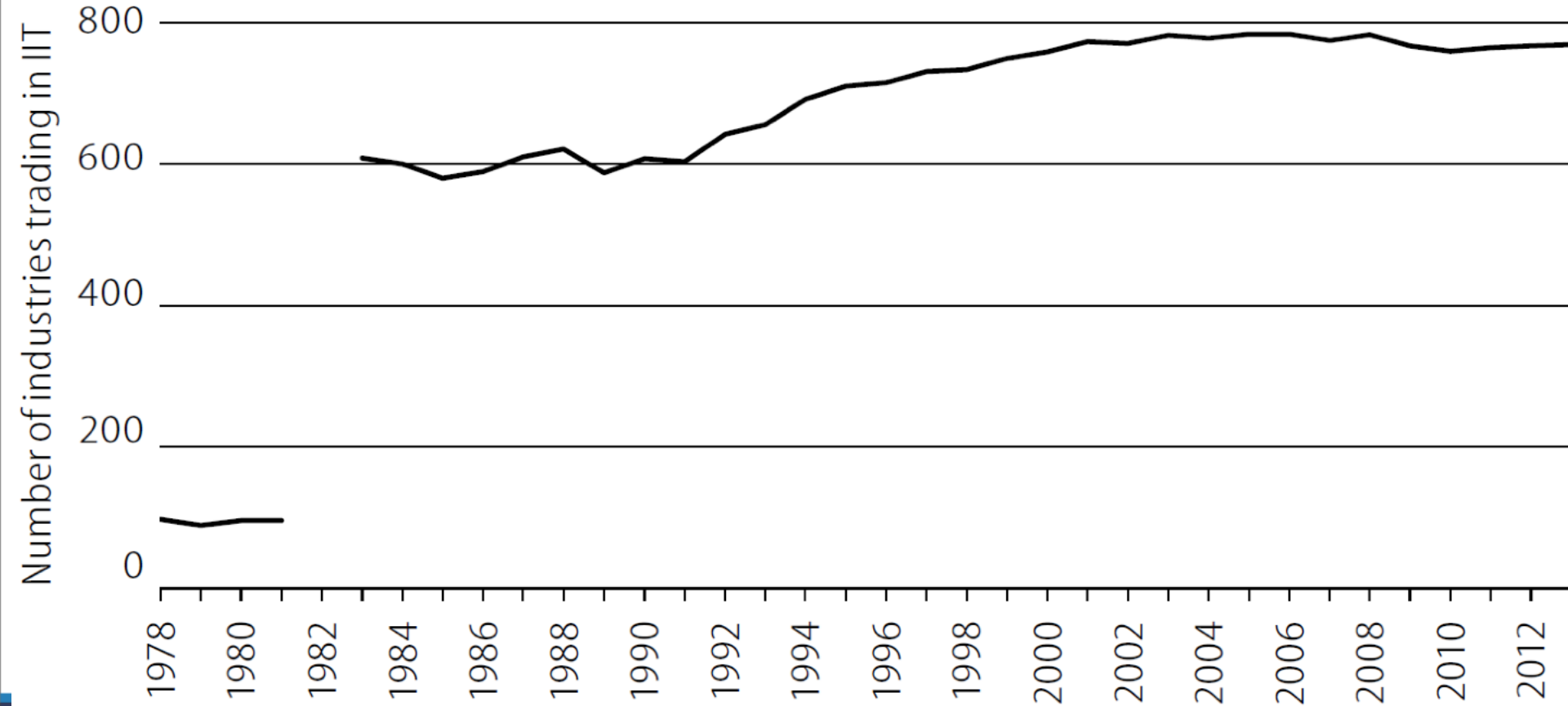
A Case: Passenger car industry

- If a product requires large volumes to be produced for the product to be profitable for a firm, then few firms will exist relative to the volume demanded. Few firms lead to less two-way trade flow of that specific product, leading to lower Horizontal IIT.
 - The passenger car industry is characterized by large minimum efficient scale, and high initial costs.
 - These factors lead to an industry with few firms.
 - Most of these firms have a differentiated product portfolio that matches the consumers demand for different varieties.
 - Even if most producers have a differentiated product portfolio, their products are often classified as belonging to a specific quality-segment.

Approaches to Measure

- Bagchi et al (2014) used the **unit value dispersion criterion** following Greenaway, Hine and Milner (**GHM**) approach at **HS-6-digit** level
- This GHM approach disentangles total IIT into horizontal and vertical, termed as, high vertical or *h-VIIT* and inferior, as, low vertical or *l-VIIT* products
- Bagchi et al (2014) estimated IIT for six major Indian manufacturing industries over the period 1990–2013.
- They found that technologically inferior quality products (or l-VIIT) have been dominant in India's export basket throughout the period of study,
- whereas HIIT and export of high technology goods (h-VIIT) have gained some momentum after the global economic slowdown in 2008

IIT in India



Data: World Bank's World Integrated Trade Solution (WITS)

Background

- most commonly used measure of IIT was proposed by **Grubel and Lloyd (1975)**
- The Grubel and Lloyd (GL) index for any particular product class i between countries A and B is defined as follows:

where, X_{AB}^i is the **value** of exports of product class i by country A to country B , and M_{AB}^i is the value of imports of product class i by country A from country B

$$GL_{AB}^i = 1 - \frac{|X_{AB}^i - M_{AB}^i|}{X_{AB}^i + M_{AB}^i}$$

- The value of this index lies between **zero and one**, including the two extreme values.

GL Measure

- **Intra-industry Trade Index (T):**

$$T = 1 - \frac{|X - M|}{X + M}$$

- X = exports
 - M = imports
 - Numerator is absolute value
 - T ranges from 0 to 1
 - $T=0$ when nation only imports or exports the good
 - $T=1$ when exports = imports.
-
- Balassa observed that trade volume of differentiated products increased within broad industrial classification. Balassa used the term IIT in 1966.
 - Grubel and Lloyd calculated the T index for 10 industrial countries in 1967 and found the range 0.30 to 0.66

GL Limitations

- Although widely used, the GL index is criticized on two grounds.
 - First, it does not take into account a country's trade imbalance.
 - second, it cannot distinguish between the type of IIT, that is, horizontal or vertical.
- To overcome the first limitation, Bergstrand (1983) proposed an index which adjusts for each country's multilateral trade imbalance in the following way:

$$BG_{AB} = 1 - \frac{|\widetilde{X}_{AB}^i - X_{BA}^i|}{\widetilde{X}_{AB}^i + X_{BA}^i}$$

$$\text{where, } \widetilde{X}_{AB}^i = \frac{1}{2} \left[\frac{X_A + M_A}{2X_A} + \frac{X_B + M_B}{2M_B} \right] X_{AB}^i \text{ and}$$

$$\widetilde{X}_{BA}^i = \frac{1}{2} \left[\frac{X_A + M_A}{2M_A} + \frac{X_B + M_B}{2X_B} \right] X_{BA}^i$$

BL Approach

- Balassa (1986) proposed the following index to measure IIT by taking the sum of the ratios of trade balance to total trade for each product group and then dividing by the number of product groups:
- The total exports by country A to country B and M_{AB} is the total imports by country A from country B .
- As the share of IIT increases, BL_{AB} declines from one to zero.

$$BL_{AB} = \frac{1 - \sum_i \left| \tilde{X}_{AB}^i + \tilde{M}_{AB}^i \right|}{\sum_i \left| \tilde{X}_{AB}^i + \tilde{M}_{AB}^i \right|}$$

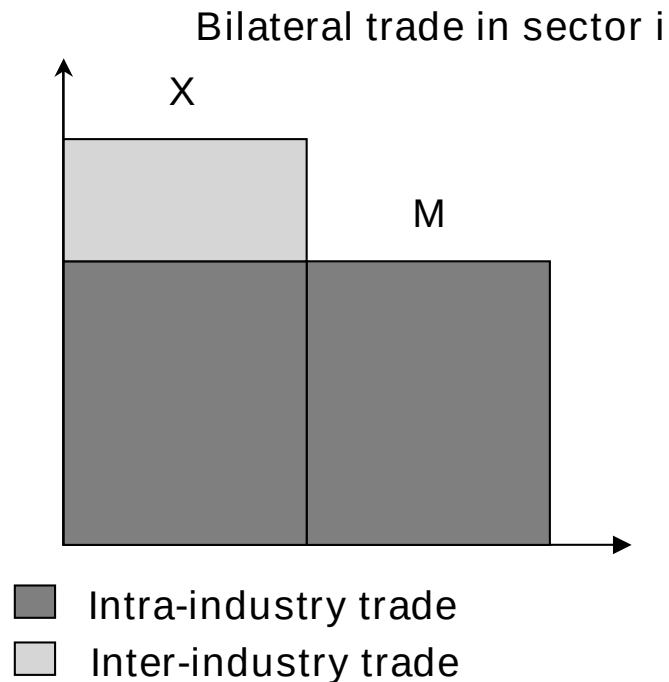
$$\text{where, } \tilde{X}_{AB}^i = \frac{X_{AB}^i [X_{AB} + M_{AB}]}{2X_{AB}}$$

$$\tilde{M}_{AB}^i = \frac{M_{AB}^i [X_{AB} + M_{AB}]}{2M_{AB}}, X_{AB}$$

- Regarding the second limitations of GL, there have been attempts to decompose IIT into horizontal and vertical differentiation.
- The first formal distinction between the two categories was by Abd-el-Rahman (1991)
- As per **Abd-el-Rahman (1991)**, if the difference between export and import unit value is maximum up to **15%**, the product will be said to be horizontally differentiated; and if this difference exceeds the 15% limit, then it would be considered as a vertically differentiated product.

Intra-industry trade – Horizontal and Vertical

- $IIT = 2 * \text{Min}(X; M)$
 - minimum of export and import is the import, therefore the IIT is twice the amount of the imports.
- **Total Trade = IntraIT + InterIT** Where
 - $\text{IntraIT} = \text{HIIT} + \text{VIIT}$
- Horizontal intra-industry trade is when imports and exports, within a specific industry during a specific time interval, are composed of products of the same quality.
 - The time interval that is studied is often a year, that is the case in this paper as well.
- Vertical IIT trade is then when the traded goods, are instead of different quality.
- Most economists assume that price reflects quality, where price is a measure of value per unit, a unit-value.
- When relative unit values are outside a specific range, that is defining what is horizontal IIT, any IIT is classified as vertical (Abd-el-Rahman (1991) approach)



GHM approach

- Based on the Abd-el-Rahman (1991) approach, **GHM** (Greenaway, Hine and Milner) define IIT, their index in the GL tradition, in product class i at the **3-digit level** of classification as:

→ → →

- The underlying assumption here is that relative prices are likely to reflect relative qualities (Stiglitz 1987).
- Hence, unit value is used as an indicator of the average price of a particular good.

$$IIT_{AB}^i = 1 - \frac{\sum_j |X_{AB}^{ij} - M_{AB}^{ij}|}{\sum_j [X_{AB}^{ij} + M_{AB}^{ij}]}$$

where, X_{AB}^{ij} is the export of product j at 5-digit level in product class i at the 3-digit level by country A to country B,
 M_{AB}^{ij} import of product j at 5-digit level in product class i at the 3-digit level by country A from country B.

HIIT

- $HIIT_{AB}^i$ is given by IIT_{AB}^i (in the right equation) for those products j , in product class i , where the unit values of exports (UVX_{ij}) and imports (UVM_{ij}) satisfy the following condition: $\rightarrow \rightarrow$
- There is no obvious value for α that is correct, in research 0.15 and 0.25 is frequently used.
- “Horizontal IIT was first defined as the simultaneous export and import of a 5 digit SITC product where the unit-value of exports (measured f.o.b.) relative to the unit-value of imports (measured c.i.f.) was within the range of $\pm 15\%$.”

$$IIT_{AB}^i = 1 - \frac{\sum_j |X_{AB}^{ij} - M_{AB}^{ij}|}{\sum_j [X_{AB}^{ij} + M_{AB}^{ij}]}$$

$$1 - \alpha \leq \frac{UVX_{ij}}{UVM_{ij}} \leq 1 + \alpha \text{ -----} 1$$

where α is an exogenously given dispersion factor.

$$UV_{EXP_{ij}} = \frac{Value_{EXP_{ij}}}{Units_{EXP_{ij}}}, UV_{IMP_{ij}} = \frac{Value_{IMP_{ij}}}{Units_{IMP_{ij}}}$$

VIIT

- On the other hand, $VIIT_{AB}^i$ is given by IIT_{AB}^i (in the right equation) for those products j in product class i for which, $\rightarrow \rightarrow$
- Although the value of α is arbitrarily chosen, 0.15 and 0.25 are the two most commonly used values in the literature

$$IIT_{AB}^i = 1 - \frac{\sum_j |X_{AB}^{ij} - M_{AB}^{ij}|}{\sum_j [X_{AB}^{ij} + M_{AB}^{ij}]}$$

$$\frac{UVX_{ij}}{UVM_{ij}} < 1 - \alpha \text{ or } \frac{UVX_{ij}}{UVM_{ij}} > 1 + \alpha$$

$$\text{if } \frac{UV_{ij}^{EXP}}{UV_{ij}^{IMP}} > 1 + \alpha$$

$$\text{or } \frac{UV_{ij}^{EXP}}{UV_{ij}^{IMP}} < 1 - \alpha$$

$$\text{then } IIT_{ij} = VIIT_{ij} \quad \text{otherwise} \quad IIT_{ij} = HIIT_{ij}$$

$$HIIT_{ij} = TT_{ij} - InterIT_{ij} - VIIT_{ij}$$

- If the difference between unit-value of exports and imports is outside the interval of $1 \pm \alpha$ then IIT is classified as vertical IIT, otherwise it is classified as horizontal IIT.
- If the trade is classified as vertical then the Grubel-Lloyd index for vertical IIT is equal to the aggregated Grubel-Lloyd index for IIT.
- This is true since at the aggregated level all IIT is either vertical or horizontal

HIIT and VIIT

- HIIT → similar in terms of quality but have different characteristics or attributes, explained by the framework developed by Dixit and Stiglitz (1977), Lancaster (1980) and Krugman (1980, 1981). Helpman and Krugman (1985) concluded that the larger is the difference in factor endowments, the smaller (larger) the extent of HIIT (VIIT).
- VIIT → trade in similar products of different qualities, which are not the same in terms unit production costs and factor intensities (Falvey, 1981; Falvey and Kierzkowski, 1987)



- Vertical Intra-industry Trade (VIIT) dominates Horizontal Intra-Industry Trade (HIIT) for the selected countries like India.
- bilateral IIT level with respect to India's high-income partners, while the same effect is non-significant for low-income countries



FF approach

- Greenaway et al (1994) calculated HIIT and VIIT for the United Kingdom
- They found VIIT to be the most important form of IIT with $\alpha = 0.15$, whereas with $\alpha = 0.25$, VIIT remains as important as HIIT.
- The second approach is by Fontagné and Freudenberg (1997) (referred to as FF hereafter) with a slight change in condition (eq1): $\rightarrow \rightarrow \rightarrow$
- This ensures symmetry between the upper and lower bounds in terms of their relative distance from unity.

$$\frac{1}{1 + \alpha} \leq \frac{UVX_{ij}}{UVM_{ij}} \leq 1 + \alpha$$

Difference Between GHM and FF

- The main difference between the GHM and the FF approach is regarding the degree of trade overlap to define IIT.
- In the FF approach, an arbitrary value $\lambda=10\%$ is taken as a cut-off implying two-way trade,
- when the degree of trade overlap is more than 10% and one way when it is less than 10%.
- But, the consequence of setting $\lambda =10\%$ means that the unit value ratios are calculated for a smaller sample of products.
- On the other side, in the Greenaway et al (1994) approach, imports and exports must exceed \$50 million to be considered as IIT, so that very small trade values are avoided.
- What this means is that use of any cut-off leads to the loss of some relevant information.
- If trade value is just slightly below the cut-off (say, 9%), even if the varieties are differentiated, such small levels of IIT will not be considered for further analysis.
- Second, the cut-offs are somewhat arbitrary. For instance, in the GHM approach, \$50 million cut-off may not be applicable for all trade relations, in particular, if such a relation involves the developing countries

AE approach

- The third approach to decompose IIT is put forward by Azhar and Elliott (2006)
- denoted as AE $\rightarrow \rightarrow \rightarrow$
- proposed the following index with symmetrical limit and country invariant nature to classify IIT into HIIT and VIIT
- When the unit value of exports exceeds that of imports or the unit value of imports exceeds that of exports by 85%, then the trade could be classified as horizontally differentiated trade.

$$PQV = 1 + \frac{UV^X - UV^M}{(UV^X + UV^M)}, \text{ with } 0 < PQV < 2$$

AE

- On the other hand, when the unit value of exports exceeds that of imports by 50%, then such trade could be classified as vertically differentiated.
- IIT is considered as VIIT when PVQ is 1.2.
- If $0.92 < PVQ < 1.08$, then it is HIIT.
- However, if $PVQ < 0.92$; $1.08 < PVQ < 2$ (excluding the value 1.2), then it is inconclusive.

AE, GHM and FF

- The main difference between the PQV index of AE and the GHM and the FF approach is that the latter two approaches use a dispersion percentile to indicate boundaries for higher or lower product quality from the perspective of home or foreign countries.
- On the other hand, the PQV index of AE does not require any such dispersion factor and is scaled and symmetric.
- But, the AE approach also has some arbitrariness and hence, an inconclusive zone.

- Grubel-Lloyd Uncorrected (GLU) formula, used for country j for industry i as the following:
- However, when the GLU index is applied for measuring the IIT between developed and developing countries, possibility of underestimation cannot be ruled out due to trade imbalance.
- The Grubel-Lloyd Corrected (GLC) formula involving country j for industry i , uses the following formulation:

$$GLU = \frac{\sum_i (X_{ij} + M_{ij}) - \sum_i |X_{ij} - M_{ij}|}{\sum_i (X_{ij} + M_{ij})} \times 100$$

$$GLC = \frac{\sum_i (X_{ij} + M_{ij}) - \sum_i |X_{ij} - M_{ij}|}{\sum_i (X_{ij} + M_{ij}) - \left| \sum_i X_{ij} - \sum_i M_{ij} \right|} \times 100$$



Figure 1: India's Overall IIT with ROW (2001-2015)

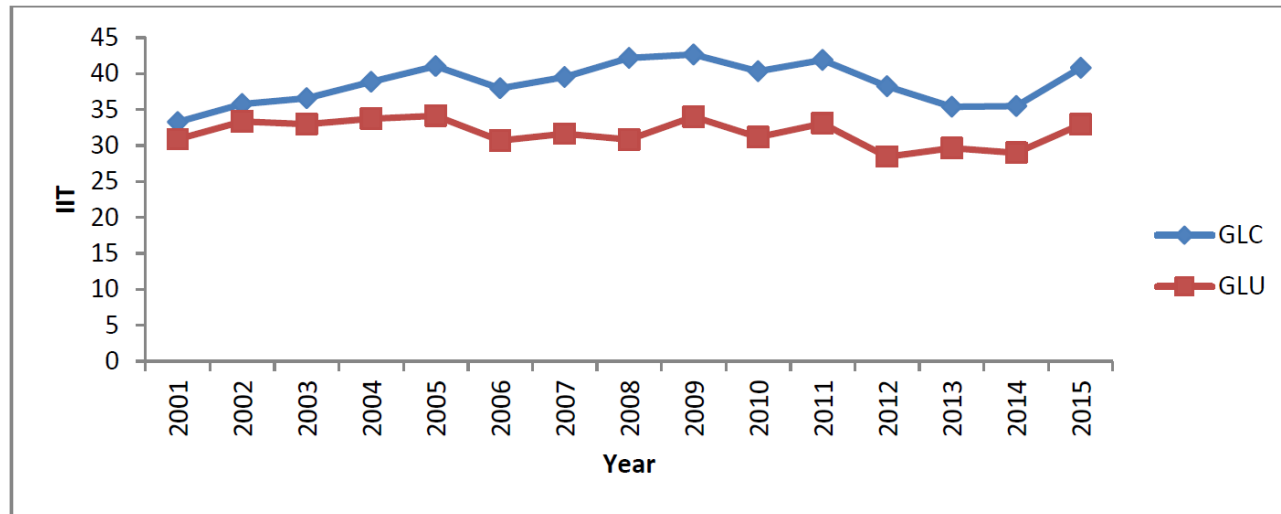


Table 2: Average Shares of India's Major Trade Partners in the Trade Basket

No.	Country	Export Share (%)			Import Share (%)		
		2001-05	2006-10	2011-15	2001-05	2006-10	2011-15
1	Australia	0.90	0.75	0.87	2.93	3.63	2.49
2	USA	18.50	12.51	12.98	6.38	6.41	4.87
3	China	4.41	6.46	4.67	5.30	10.66	12.52
4	Indonesia	1.47	1.61	1.67	2.26	2.38	3.20
5	Japan	2.97	2.11	1.95	3.22	2.53	2.37
6	Korea	1.24	1.89	1.43	2.86	2.76	2.88
7	Iran	1.14	1.22	1.18	0.43	3.80	2.28
8	South Africa	1.01	1.46	1.62	2.51	1.70	1.63
9	UK	4.78	3.80	3.06	4.11	1.88	1.35
10	Qatar	0.20	0.29	0.30	0.34	1.29	3.00
11	Malaysia	1.43	1.53	1.50	2.23	2.24	2.19
12	Thailand	1.25	1.06	1.12	0.77	0.98	1.22
13	Sri Lanka	1.77	1.51	1.66	0.24	0.19	0.15
14	Germany	3.82	3.14	2.54	3.83	3.81	2.97
15	Switzerland	0.73	0.36	0.40	4.91	4.71	5.59
16	Netherland	2.06	3.05	2.64	0.77	0.66	0.55
17	Singapore	3.53	4.46	4.02	2.47	2.62	1.66
18	Hong Kong	4.95	3.96	4.24	1.60	1.69	1.73
19	Vietnam	0.62	0.95	1.65	0.06	0.15	0.53
20	Bangladesh	2.22	1.43	1.73	0.09	0.11	0.13
21	Brazil	3.02	2.48	2.03	0.57	0.66	1.00
22	Belgium	0.72	1.40	1.83	4.90	2.08	2.22
23	Italy	2.66	2.27	1.63	1.34	1.40	1.02
24	Nigeria	0.93	0.79	0.89	0.10	2.93	2.96
25	France	2.08	1.85	1.70	1.55	1.68	0.80
	Total	68.41	62.34	59.31	55.77	62.95	61.31



Table 3: India's IIT Results for Top Trade Partners

	Intra Industry Trade Index			Partnership / Negotiations with India through Trade Bloc	Status
Country	2001-05	2006-10	2011-15		
Developed Economies					
Australia	11.25	12.84	7.69	CECA, RCEP	Under Negotiations
Belgium	62.88	50.72	51.98	India-EU BTIA	Under Negotiations
France	19.42	22.82	33.19	India-EU BTIA	Under Negotiations
Germany	25.57	35.39	40.10	India-EU BTIA	Under Negotiations
Hong Kong, SAR	60.05	64.29	57.84		No FTA
Italy	27.85	24.12	30.87	India-EU BTIA	Under Negotiations
Japan	13.03	18.05	19.56	JICEPA, RCEP	CEPA
Netherlands	23.91	24.88	25.49	India-EU BTIA	Under Negotiations
Qatar	1.22	7.11	15.11	GCC	Framework Agreement signed
Singapore	21.19	48.41	39.44	ISCECA, IASEAN FTA, RCEP	FTA, CECA
South Korea	17.71	29.90	38.03	IKCEPA, RCEP	CEPA
Switzerland	36.86	43.54	36.10	India-EFTA Agreement	Under Negotiations
UK	18.22	25.85	27.53	India-EU BTIA	Under Negotiations
USA	31.21	26.63	29.82		No FTA
Developing Economies					
Bangladesh	12.66	16.85	22.74	SAFTA, BIMSTEC	FTA
Brazil	6.69	10.28	7.76	India Mercosur PTA, IBSA	PTA
China	15.07	15.12	20.36	APTA, RCEP	PTA
Indonesia	11.74	14.36	13.22	IICECA, IASEAN FTA, RCEP	FTA
Iran	9.00	9.35	3.43	GSTP	No FTA
Malaysia	19.03	22.63	24.01	IMCECA, IASEAN FTA, RCEP	CECA
Nigeria	7.56	0.53	0.47	GSTP	No FTA
South Africa	4.97	5.64	3.97	IBSA, SACU PTA	Under Negotiations
Sri Lanka	29.80	30.77	43.33	ISLFTA, BIMSTEC	FTA
Thailand	20.35	25.61	30.29	BIMSTEC, IASEAN FTA, RCEP	FTA
Vietnam	10.55	16.04	12.87	IASEAN FTA, RCEP	FTA



*Thank
you*

